

Business Analysis Capstone Project

Customer Funnel Optimization:

Analysis of Customer Behavior, Conversion Performance, and Revenue Drivers

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Tools Used: Microsoft Excel | MySQL | Power BI

Data Source

The dataset used in this project was obtained from Kaggle and contains ecommerce customer funnel data, including session activity, cart creation, purchases, revenue, marketing channels, customer segments, and engagement metrics.

- ☐ Kaggle Dataset
- ☐ Ecommerce.CSV
- ☐ File Used: ecommerce.csv

Project Overview

1. Project Title

Customer Funnel Optimization: Analysis of Customer Behavior, Conversion Performance, and Revenue Drivers Using Excel, SQL, and Power BI

2. Background

E-commerce businesses invest significant resources in attracting customers to their platforms; however, not all visitors complete a purchase. Understanding customer behavior across the purchase funnel is critical for identifying conversion bottlenecks, reducing cart abandonment, and maximizing revenue. This project analyzes customer interactions across multiple stages of the ecommerce funnel to uncover patterns influencing conversion performance and business outcomes.

3. Project Objective

The objective of this project is to analyze customer funnel performance and identify opportunities to improve conversion rates and revenue generation. Using Excel for exploratory data analysis, SQL for business analysis, and Power BI for visualization, the project investigates customer behavior, marketing channel effectiveness, product category performance, and the impact of customer engagement on purchasing decisions.

Excel Analysis

1. Objective

Exploratory Data Analysis (EDA) was conducted using Microsoft Excel to understand the structure, quality, and distribution of the ecommerce dataset. The objective was to identify key customer funnel metrics, validate data consistency, detect potential anomalies, and establish a foundation for deeper analysis using SQL and Power BI.

2. Dataset Overview

The dataset contains customer interaction data from an e-commerce platform. It captures user sessions, cart activity, purchases, revenue generation, customer engagement, marketing channels, device usage, and product category information. The dataset was used to analyze customer funnel performance and identify opportunities for conversion optimization.

2.2 Dataset Size:

Metric	Value
Total Records	25000
Total Columns	29
Dataset Format	CSV
Source	Ecommerce Customer Funnel Dataset

2.3 Key Variables

Column Name	Description
session_id	Unique customer session identifier
user_type	Customer segment classification
marketing_channel	Customer acquisition channel
device_type	Device used during session
added_to_cart	Cart creation indicator
purchased	Purchase completion indicator
revenue	Revenue generated
session_duration	Duration of customer session
product_category	Product category purchased

2.4 Data Types Summary

Data Type	Examples
Numerical	Revenue, Session Duration
Categorical	User Type, Device Type, Marketing Channel
Binary	Added To Cart, Purchased

3. Data Quality Observations

- Column names were clearly defined and easy to interpret.
- The dataset was structured and suitable for analytical processing.
- No major data integrity issues were identified during the initial review and exploratory analysis.

- Missing values, duplicate records, and data consistency were assessed as part of the data preparation process.
- The dataset was determined to be suitable for KPI calculation, SQL analysis, and dashboard development.

4. Descriptive Statistics

Objective - Descriptive statistics were used to summarize key characteristics of the ecommerce dataset and establish baseline customer funnel performance metrics before conducting deeper business analysis.

Summary Metrics

Metric	Value
Total Sessions	25,000
Add-to-Cart Actions	16,117
Purchases	5,616
Conversion Rate	22.46%
Cart Abandonment Rate	65.15%
Total Revenue	10M+

Key Observations

- The dataset contains **25,000 customer sessions**, providing a substantial sample for funnel analysis.
- Approximately **64% of sessions resulted in a cart creation**, indicating strong initial purchase intent.
- Only **22.46% of sessions resulted in a completed purchase**, highlighting opportunities for conversion optimization.
- The **cart abandonment rate of 65.15%** indicates significant customer drop-off between cart creation and purchase completion.
- The business generated over **10 million in revenue**, demonstrating the commercial significance of funnel performance improvements.

Conclusion

The descriptive statistics indicate that customer interest is relatively strong at the cart creation stage; however, a significant proportion of customers do not complete their purchases. The high abandonment rate suggests potential friction within the purchase journey and justifies further investigation through SQL-based business analysis and dashboard visualization.

5. Funnel Overview

The customer funnel consists of three primary stages: Session, Add-to-Cart, and Purchase. Analysis of 25,000 customer sessions revealed that 16,117 sessions progressed to the cart stage, while only 5,616 resulted in completed purchases. Although customer interest remained strong during product exploration, a substantial reduction was observed between cart creation and purchase completion, indicating potential opportunities for conversion optimization.

6. Initial Observations

- Customer acquisition appears healthy, with a large volume of sessions entering the funnel.
- A significant number of users demonstrate purchase intent by adding products to their carts.
- The largest performance gap occurs between the cart and purchase stages.
- Conversion opportunities are likely influenced by customer behavior, acquisition channels, product preferences, and engagement levels.
- Further analysis is required to identify the factors contributing to funnel leakage and revenue performance.

SQL Analysis

1.Objective

SQL was used to perform diagnostic analysis on customer funnel performance. While Excel helped identify what happened and Power BI visualized the results, SQL was used to investigate why funnel leakage occurred and identify key drivers of conversion, abandonment, and revenue generation.

2.Analysis Framework

Business Question	Purpose
Which customer segment abandons the most?	Identify high-risk customer groups
Which marketing channels perform best?	Evaluate acquisition effectiveness
Which product categories generate the most revenue?	Understand revenue drivers
Does customer engagement impact purchases?	Measure behavioral influence on conversion

3.1: Customer Segment Abandonment

Business Question - Which customer segment abandons the funnel most frequently?

Objective: Understanding abandonment behavior across customer segments helps prioritize retention strategies and improve conversion rates.

Query

```
/*=====
SQL ANALYSIS 1
Which customer segment abandons carts the most?
=====*/
SELECT
  user_type,
  COUNT(*) AS total_sessions,

  SUM(CASE
    WHEN added_to_cart = 1
    THEN 1
    ELSE 0
  END) AS carts_created,

  SUM(CASE
    WHEN added_to_cart = 1
      AND purchased = 0
    THEN 1
    ELSE 0
  END) AS abandoned_carts,
```

```

-- Added CONCAT to match your exact expected layout with the % sign
CONCAT(
    ROUND(
        SUM(CASE
            WHEN added_to_cart = 1
            AND purchased = 0
            THEN 1
            ELSE 0
        END)
        * 100.0
        /
        NULLIF(
            SUM(CASE
                WHEN added_to_cart = 1
                THEN 1
                ELSE 0
            END),
            0
        ),
        2
    ),
    '%'
) AS abandonment_rate

FROM ecommerce_db.ecommerce

GROUP BY user_type

ORDER BY abandonment_rate DESC;

```

Result

Result Grid Filter Rows: Export: Wrap Cell Content:					
	user_type	total_sessions	carts_created	abandoned_carts	abandonment_rate
▶	0	11218	7254	5173	71.31%
	1	13782	8863	5328	60.12%

Key Findings

- User Type 0 recorded the highest cart abandonment rate at **71.31%**.
- User Type 1 recorded a lower abandonment rate of **60.12%**.
- Although User Type 1 generated more sessions and carts,they converted more effectively than User Type 0.
- The gap of over **11 percentage points** indicates a significant behavioral difference between the two customer segments.

Business Insight

- The business should prioritize retention and conversion strategies for User Type 0 customers. Potential initiatives include personalized offers, checkout optimization, targeted remarketing campaigns, simplified payment processes, and trust-building measures such as customer reviews and secure payment messaging. Monitoring the conversion behavior of this segment can help identify the specific causes of abandonment and improve overall funnel efficiency.

2: Marketing Channel Performance

Business Question - Which marketing channels perform best?

Objective - Explain why channel effectiveness matters.

Query

```
/*=====
SQL ANALYSIS 2

Which marketing channels perform best?

=====*/

SELECT
marketing_channel,
COUNT(*) AS total_sessions,
SUM(CASE
    WHEN purchased = 1
    THEN 1
    ELSE 0
END) AS purchases,
CONCAT(
    ROUND(
        SUM(CASE WHEN purchased = 1 THEN 1 ELSE 0 END) * 100.0 / COUNT(*),
        2
    ),
    '%'
) AS conversion_rate,
ROUND(SUM(revenue), 2) AS total_revenue,
-- Professional Business Metric: True Revenue Per Session (RPS)
```



```

ROUND(SUM(revenue) / COUNT(*), 2) AS avg_revenue_per_session

FROM ecommerce_db.ecommerce

GROUP BY marketing_channel

ORDER BY total_revenue DESC;

```

Result

	marketing_channel	total_sessions	purchases	conversion_rate	total_revenue	avg_revenue_per_session
5		4263	1006	23.60%	1820823.79	427.12
2		4281	943	22.03%	1711962.69	399.9
3		4116	932	22.64%	1675463.47	407.06
0		4041	918	22.72%	1661328.76	411.12
1		4190	927	22.12%	1646768.68	393.02
4		4109	890	21.66%	1599821.67	389.35

Key Findings

- Channel 5 delivered the strongest performance with a **23.60% conversion rate**, the highest total revenue (**1,820,823.79**) and the highest average revenue per session (**427.12**).
- Channel 3 achieved the second-highest conversion rate (**22.64%**) and maintained strong revenue generation.
- Channels 0, 1, 2, and 3 performed relatively similarly, with conversion rates ranging between **22.03% and 22.72%**.
- Channel 4 was the weakest-performing channel, generating the lowest conversion rate (**21.66%**), lowest total revenue (**1,599,821.67**), and lowest average revenue per session (**389.35**).
- The performance gap between the best and worst channels suggests that acquisition source has a measurable impact on customer conversion and revenue outcomes.

Business Insight

- The business should consider increasing marketing spend and campaign optimization efforts on high-performing channels such as Channel 5, which demonstrates superior conversion efficiency and revenue generation. Simultaneously, a detailed review of Channel 4 should be conducted to identify potential issues related to audience targeting, campaign messaging, traffic quality, or user experience. Continuous monitoring of channel-level performance can support data-driven budget allocation and improve overall return on marketing investment.

3: Revenue by Product Category

Business Question - Which product categories generate the most revenue?

Objective - Explain revenue optimization purpose.

Query:/*=====

SQL ANALYSIS 3

Which Product Categories Generate the Most Revenue?

=====*/

SELECT

```
product_category,

COUNT(*) AS total_transactions,

SUM(quantity) AS total_units_sold,

-- Option: Added CONCAT to attach the ₹ sign if you want it straight from SQL

CONCAT('₹', FORMAT(SUM(CASE WHEN purchased = 1 THEN revenue ELSE 0 END), 0)) AS total_revenue,

CONCAT('₹', FORMAT(ROUND(AVG(CASE WHEN purchased = 1 THEN revenue END), 2), 0)) AS avg_order_value

FROM ecommerce_db.ecommerce

GROUP BY product_category

ORDER BY SUM(CASE WHEN purchased = 1 THEN revenue ELSE 0 END) DESC;
```

Result:

Result Grid   Filter Rows: Export:  Wrap Cell Content: 					
	product_category	total_transactions	total_units_sold	total_revenue	avg_order_value
▶	2	3237	8019	₹2,037,577	₹2,826
	6	3112	7765	₹1,930,885	₹2,514
	5	3149	7949	₹1,729,167	₹2,375
	3	3040	7582	₹1,293,650	₹2,050
	7	3073	7688	₹1,089,618	₹1,570
	0	3142	7730	₹880,077	₹1,257
	1	3043	7534	₹702,789	₹1,086
	4	3204	7959	₹452,407	₹622

Key Findings

- **Category 2** generated the highest revenue (**₹2.04M**) and highest average order value (**₹2,826**), making it the strongest revenue contributor.
- **Categories 6 and 5** were also top-performing categories, each generating over **₹1.7M** in revenue.
- **Category 4** generated the lowest revenue (**₹452K**) and lowest average order value (**₹622**) despite having transaction volumes comparable to other categories.
- Revenue differences are driven more by **average order value** than by transaction volume, as most categories recorded similar transaction counts.
- Categories **2, 5, and 6** collectively represent the most valuable product segments in the business.

Business Insight

- Prioritize marketing, inventory planning, and promotional efforts for Categories 2, 5, and 6 to maximize revenue generation. Conduct a detailed review of Category 4 to identify opportunities for improving pricing strategy, product mix, or customer perception and increase its revenue contribution.

4: Impact of Customer Engagement on Purchases

Business Question - Does customer engagement impact purchases?

Objective - Explain why engagement matters.

Query:

/*=====

SQL ANALYSIS 4

Does Customer Engagement Impact Purchases?

=====*/

SELECT

session_duration_bucket,

COUNT(*) AS total_sessions,

SUM(CASE

WHEN purchased = 1

THEN 1

ELSE 0

END) AS purchases,

```

CONCAT(

ROUND(

SUM(CASE

WHEN purchased = 1

THEN 1

ELSE 0

END) * 100.0

/ COUNT(*),

2

),

'%') AS conversion_rate

FROM ecommerce_db.ecommerce

GROUP BY session_duration_bucket

ORDER BY ROUND(SUM(CASE WHEN purchased = 1 THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) DESC;

```

Result

Result Grid Filter Rows: Export: Wrap Cell Content:				
	session_duration_bucket	total_sessions	purchases	conversion_rate
▶	Long	6254	1477	23.62%
	Very Long	6240	1449	23.22%
	Short	6236	1421	22.79%
	Very Short	6270	1269	20.24%

Key Findings

- Customers with **Long sessions** recorded the highest conversion rate (**23.62%**), indicating stronger purchase intent and engagement.
- Short** and **Very Long** sessions achieved similar conversion rates (**22.79%–23.22%**), suggesting that increased session duration generally supports conversion.

- **Very Short sessions** produced the lowest conversion rate (**20.24%**), making them the least likely to result in a purchase.
- The difference between the highest and lowest conversion rates is approximately **3.4 percentage points**, demonstrating a positive relationship between customer engagement and purchasing behavior.
- Overall, customers who spend more time interacting with the platform tend to convert at higher rates.

Business Insight

- Focus on increasing session engagement by improving website navigation, product discovery, personalized recommendations, and customer interaction opportunities. Encouraging users to spend more time on the platform can help increase conversion rates and reduce funnel leakage.

5. SQL Analysis Summary

Create a summary table:

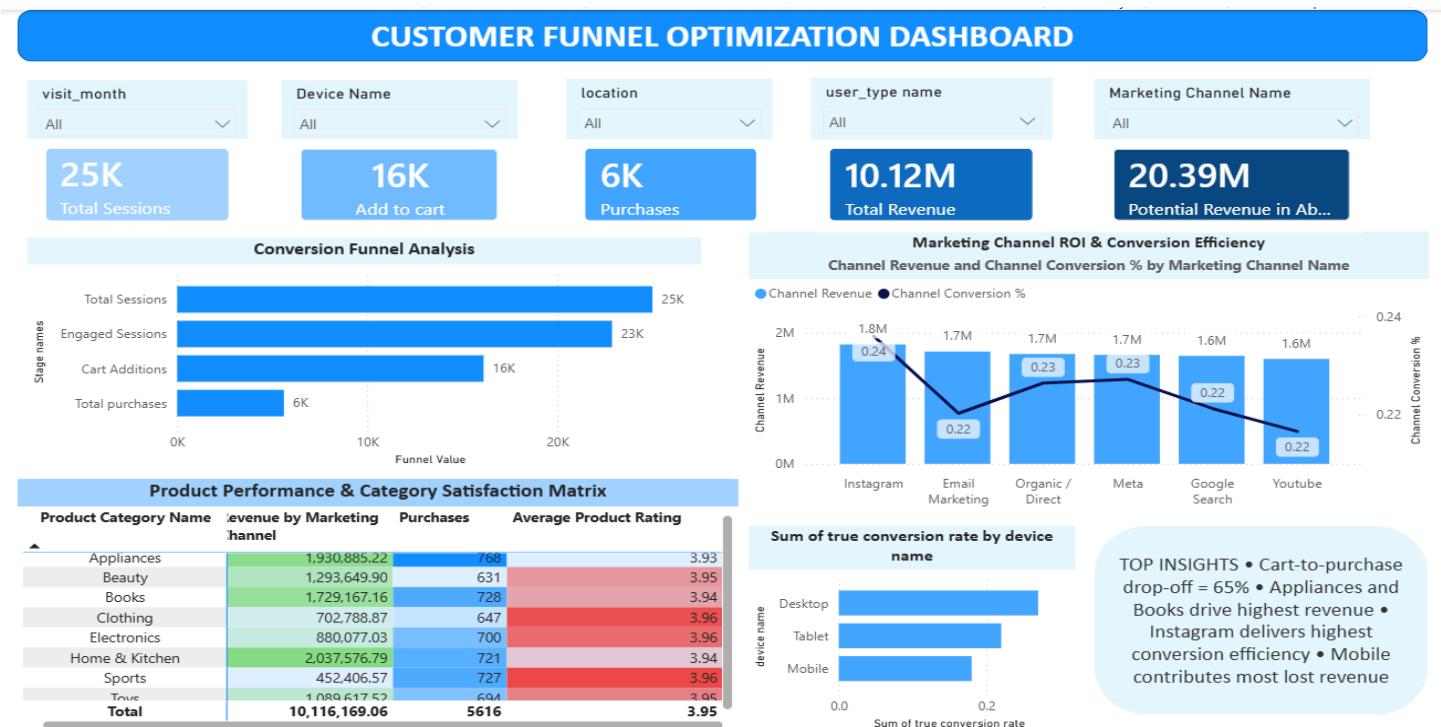
Analysis	Key Insight	Business Impact
Customer Segment	Highest abandonment identified	Retention opportunity
Marketing Channel	Best-performing channels identified	Marketing optimization
Product Category	Revenue drivers identified	Revenue growth
Engagement	Engagement influences purchases	Conversion improvement

Power BI Analysis

1. Dashboard Objective

The Power BI dashboard was developed to provide stakeholders with a consolidated analytics and business view of customer funnel performance, marketing effectiveness, customer behavior, and product performance. The objective was to identify revenue leakage, understand conversion patterns, and uncover opportunities for business growth.

2. Dashboard Screenshot



3. KPI Definitions

KPI	Definition
Conversion Rate	Purchases / Sessions
Cart Abandonment Rate	Abandoned Carts / Add to cart
Revenue	Total revenue generated
Average order value	Revenue / Purchases
Abandoned Cart Revenue Opportunity	Potential revenue associated with abandoned carts

4. Dashboard Insights

Executive Overview

- Revenue generated: 10.12M
- Revenue opportunity in abandoned carts: 20.39M
- Conversion rate indicates substantial opportunity to improve customer purchase completion.

Funnel Analysis

- Largest drop-off occurs between Add-to-Cart and Purchase
- Approximately 65% of users who added products to their cart did not complete a purchase.

Marketing Performance

- Channel performance varies significantly
- Certain channels generate higher traffic but weaker conversion efficiency
- Budget optimization opportunities exist

Customer Behavior

- Device and engagement patterns influence conversion outcomes
- Mobile users represent a key opportunity segment for further investigation

Product Performance

- Revenue is concentrated within a limited number of product categories
 - Category performance varies by purchases, ratings, and discount levels
-

5. Business Recommendations

Recommendation 1 - Investigate and address factors contributing to cart abandonment to improve purchase completion rates.

Recommendation 2 - Investigate mobile-user conversion behavior to identify potential friction points.

Recommendation 3 - Prioritize marketing investment toward higher-performing channels.

Recommendation 4 - Focus inventory and promotional efforts on top-performing product categories.

6. Business Impact

Expected Business Impact

- Reducing cart abandonment can unlock a significant portion of the 20.39M revenue opportunity identified in abandoned carts.
- Optimizing high-performing marketing channels may improve marketing ROI and revenue generation.
- Improving conversion performance among mobile users could increase overall purchase rates.
- Focusing on top-performing product categories can support revenue growth and inventory optimization.